This assignment will not be graded and is only for practice.

Level 1

1) Find the rank of the following matrix: $\begin{bmatrix} 0 & 1 & 3 & 1 \\ 1 & 0 & 0 & 1 \\ 3 & 2 & 0 & 3 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Consider the following set of vectors S in $\mathbb{R}^3$ to answer the questions 2, 3 and 4.
$S = \{(1, 2, 0), (0, 3, 1), (3, 3, -1), (3, 0, -2)\}$.

2) Let V be the vector space spanned by the vectors $(1, 2, 0)$ and $(0, 3, 1)$. Which one of the following is correct? ***1 point***

- $V = \{(a, 2a + 3b, b) \mid a, b \in \mathbb{R}\}$
- $V = \{(0, 2a - 3b, b) \mid a, b \in \mathbb{R}\}$
- $V = \{(-a, 2a + 3b, 0) \mid a, b \in \mathbb{R}\}$
- $V = \{(0, 2a + 3b, 0) \mid a, b \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$V = \{(a, 2a + 3b, b) \mid a, b \in \mathbb{R}\}$

3) If the vectors in S are written as the columns of a matrix A, then what will be the rank of A?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) What will the dimension of the vector space spanned by S be?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Choose the set of correct options from the following. ***1 point***

- [ ] Row rank and column rank of a matrix is always the same.
- [ ] The rank of a zero matrix is always 0.
- [ ] The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.
- [ ] Rank of the $n \times n$ identity matrix is 1 for any $n \in \mathbb{N}$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Row rank and column rank of a matrix is always the same.
The rank of a zero matrix is always 0.

The rank of a matrix, all of whose entries are the same non-zero real number, must be 1.

Level 2

6) Consider the following upper triangular matrix to choose the correct options. ***1 point***

$$\begin{bmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{bmatrix}$$

where $a, b, c, d, e, f \in \mathbb{R}$.

- [ ] If $f = 0$, then the rank of the matrix must be less than or equal to 2.
- [ ] If $f = 0$, then the rank of the matrix must be exactly 2.
- [ ] If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
- [ ] If a, d, f are non-zero then the rank of the matrix must be 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $f = 0$, then the rank of the matrix must be less than or equal to 2.
If a, b, c, d, e, f are all non-zero then the rank of the matrix must be 3.
If a, d, f are non-zero then the rank of the matrix must be 3.

7) If rank of the matrix $\begin{bmatrix} 2 & -3 & 4 \\ 0 & 1 & -2 \\ 1 & -3 & a \end{bmatrix}$ is 2 then find the value of a.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

8) Find the rank of the matrix A, where $A = [a_{ij}]$ is of order 3×3 and $a_{i,j} = \min\{i, j\}, i, j = 1, 2, 3.$
[Hint: Write down the matrix and apply row operations to derive it's reduced row echelon form.]

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

9)
What is the rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{bmatrix}$, where x, y, z are distinct non-zero integers?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

10) Rank of a 3×3 matrix A is 3 and another 3×3 matrix B is 2. What is the minimum possible value for the rank of AB?

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Your score is: 0/10